✍️Mastering Vim: Command Notes & Examples

Vim (**Vi IMproved**) is a powerful text editor for Linux/Unix. It's based on **vi (Visual Interface)** but with many improvements. Unlike typical editors, Vim works in **modes** (Normal, Insert, Command-line, Visual), and most editing is done using **keyboard shortcuts**, not the mouse.

This guide will help you **learn and master Vim** step by step with clear explanations, commands, and outputs.

👃 Starting Vim

```
$ vim filename.txt
```

- Opens `filename.txt` in Vim. - Creates the file if it doesn't exist.

💘Vim Modes

1. **Normal Mode** (default)
2. Used for navigation, deleting, copying, pasting.
3. Press `Esc` to return here anytime.
4. **Insert Mode** (typing mode)
5. Used for entering text.
6. Enter using:
    - `i` → insert before cursor
    - `I` → insert at beginning of line
    - `a` → append after cursor
    - `A` → append at end of line
    - `o` → open new line below
    - `O` → open new line above
7. Exit Insert Mode: `Esc`.
8. **Command-line Mode**
9. Used for saving, quitting, searching, replacing.
10. Enter using `:` (colon) in Normal Mode.
11. Examples: `:w` (save), `:q` (quit).
12. **Visual Mode** (selection)
13. Used for selecting blocks of text.

14. Enter with `v` (character), `V` (line), `Ctrl+v` (block).
15. Then apply commands like delete (`d`) or copy (`y`).

Navigation in Normal Mode

Key	Action
`h`	Move left
`l`	Move right
`j`	Move down
`k`	Move up
`0`	Move to beginning of line
`$`	Move to end of line
`w`	Jump forward one word
`b`	Jump backward one word

Think of `h j k l` like arrow keys under your right hand.

Example text:

```
Hello World
```

- Cursor on `H`, press `w` → jumps to `W`. - Cursor on `H`, press `$` → jumps to `d` (end of line).

✍️Insert Text

Example text:

```
Hello World
```

Commands:

```
i   → Insert before cursor
I   → Insert at beginning of line
a   → Insert after cursor
A   → Insert at end of line
o   → Open new line below and start insert
O   → Open new line above and start insert
```

Cursor on `W`: - Press `i` → you can type before `W`. - Press `a` → you can type after `W`. - Press `o` → new line opens below, ready to type.

🟫 Save & Quit

In Normal Mode, press `:` to enter Command-line Mode:

Command	Action
`:w`	Save file
`:q`	Quit
`:wq`	Save and quit
`:q!`	Quit without saving
`:wq!`	Force save and quit

Example:

```
:wq
```

Saves changes and exits Vim.

🤩 Deleting Text

Command	Action
`x`	Delete character under cursor
`dw`	Delete word from cursor
`dd`	Delete current line
`d$`	Delete from cursor to end of line

Example text:

```
Hello World
```

- Cursor on `W`, type `dw` → removes `World`. - Cursor anywhere on line, type `dd` → removes whole line.

😏Undo & Redo

Command	Action
`u`	Undo last change
`Ctrl+r`	Redo undone change

👃 Copy, Cut & Paste

Command	Action
`yy`	Yank (copy) a line
`p`	Paste after cursor
`P`	Paste before cursor
`dd`	Cut (delete) a line

Example: - Cursor on line → `yy` → copies it. - Move elsewhere → `p` → pastes copy below.

👉🏾Searching in Files

Command	Action
`/word`	Search forward
`?word`	Search backward
`n`	Next occurrence
`N`	Previous occurrence

Example:

```
/search
```

Highlights `search` moving forward.

😌Replacing Text

Command	Action
`rX`	Replace character under cursor with `X`
`:%s/old/new/g`	Replace all `old` with `new`

Command	Action
`:%s/old/new/gc`	Replace all with confirmation

Example text:

```
Hello World
Hello World
```

```
:%s/World/Linux/g
```

Output:

```
Hello Linux
Hello Linux
```

👂 Working with Multiple Files

```
$ vim file1.txt file2.txt
```

Switch files: - `:next` → next file - `:prev` → previous file

🤓Splitting Windows

Command	Action
`:split filename`	Horizontal split
`:vsplit filename`	Vertical split
`Ctrl+w w`	Switch between windows

🖕 Force Quit

```
:q!
```

Quit without saving.

👹Summary Cheatsheet

- **Modes**: Normal (Esc), Insert (i, a, o), Command-line (:), Visual (v)
- **Save/Quit**: `:w`, `:q`, `:wq`, `:q!`
- **Navigation**: `h j k l`, `w`, `b`, `0`, `$`
- **Edit**: `i`, `a`, `o`, `O`
- **Delete**: `x`, `dw`, `dd`, `d$`
- **Undo/Redo**: `u`, `Ctrl+r`
- **Copy/Paste**: `yy`, `p`, `P`
- **Search**: `/word`, `?word`, `n`, `N`
- **Replace**: `:%s/old/new/g`
- **Multiple Files**: `:next`, `:prev`
- **Windows**: `:split`, `:vsplit`, `Ctrl+w w`

✅With these commands and examples, you'll be able to **navigate, edit, search, and manage files in Vim efficiently**.